

NLP Laboratory — Assignment 5

Add-k Smoothing ($k = 0.3$) for N-gram Language Models

Name	_____	Course	Natural Language Processing
Roll No.	_____	Lab	Lab 5 — Add-k Smoothing

222,199
Vocabulary size (V)

0.3
Smoothing constant k

4
N-gram orders evaluated

3.10%
Test OOV rate

This lab reuses the four n-gram language models (unigram, bigram, trigram, quadrigram) trained on the Gujarati corpus in Assignment 4, and rescores them with **Add-k smoothing** instead of Add-one (Laplace) smoothing, on the same held-out dev and test splits.

1. Overview

Add-k smoothing generalizes Laplace (add-one) smoothing by replacing the fixed "+1" with a tunable constant k, redistributing less probability mass to unseen n-grams than Laplace does. All counts (n-gram counts, context counts, vocabulary, and the <UNK> mapping) are identical to Assignment 4 — only the scoring formula changes — so the two labs' perplexities are directly comparable.

$$P_{\text{add-k}}(w_i | w_{i-N+1} \dots w_{i-1}) = \frac{C(w_{i-N+1} \dots w_i) + k}{C(w_{i-N+1} \dots w_{i-1}) + k \cdot V}$$

with $k = 0.3$ and $V = 222,199$ (kept training words + <UNK> + </s>, singleton words collapsed to <UNK>).

2. Results

Order	Laplace dev PP	Add-k dev PP	Laplace test PP	Add-k test PP	Test improvement
Unigram	2,610.9	2,605.6	2,429.3	2,422.6	+0.3%
Bigram	5,359.8	2,976.9	4,896.4	2,725.9	+44.3%
Trigram	37,653.8	25,575.9	34,912.7	23,594.5	+32.4%
Quadrigram	87,766.2	70,423.6	83,181.8	66,251.2	+20.4%

Add-k ($k=0.3$) reduces perplexity relative to Laplace ($k=1$) at every order, on both dev and test — the green improvement column is positive throughout.

3. Comparison Charts

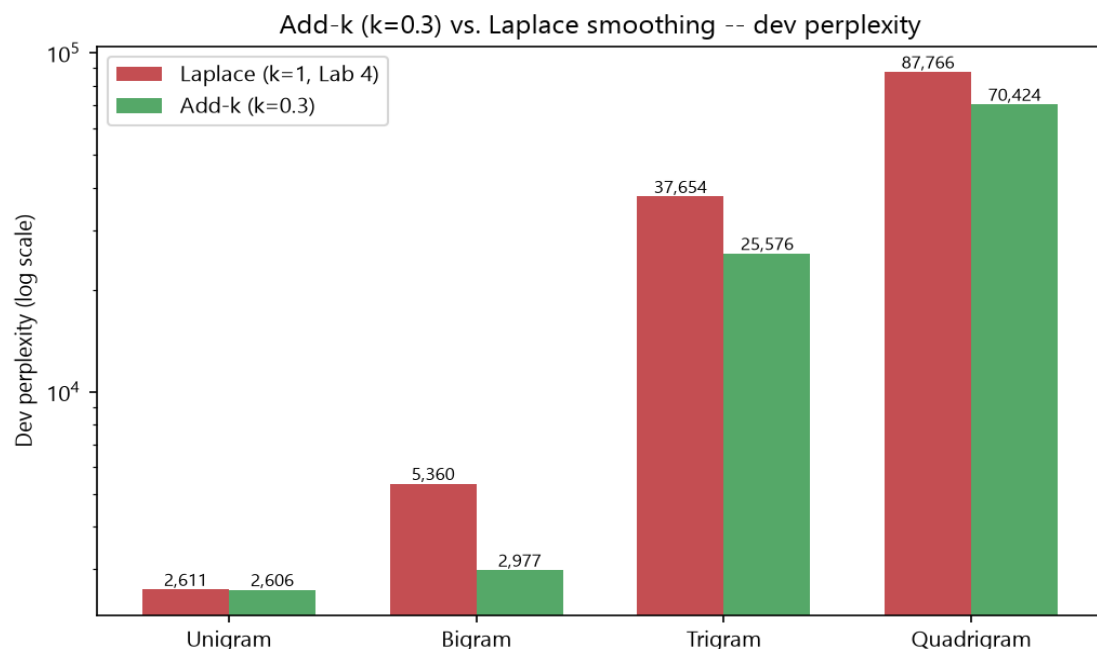


Figure 1 — Dev-set perplexity, Add-k vs. Laplace, by n-gram order.

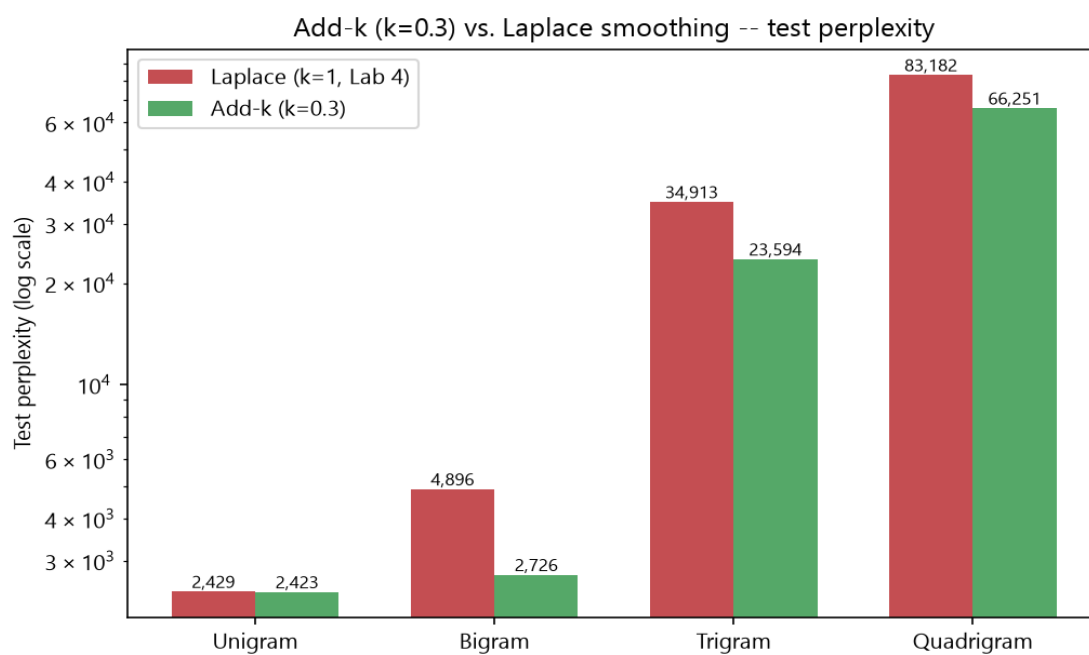


Figure 2 — Test-set perplexity, Add-k vs. Laplace, by n-gram order.

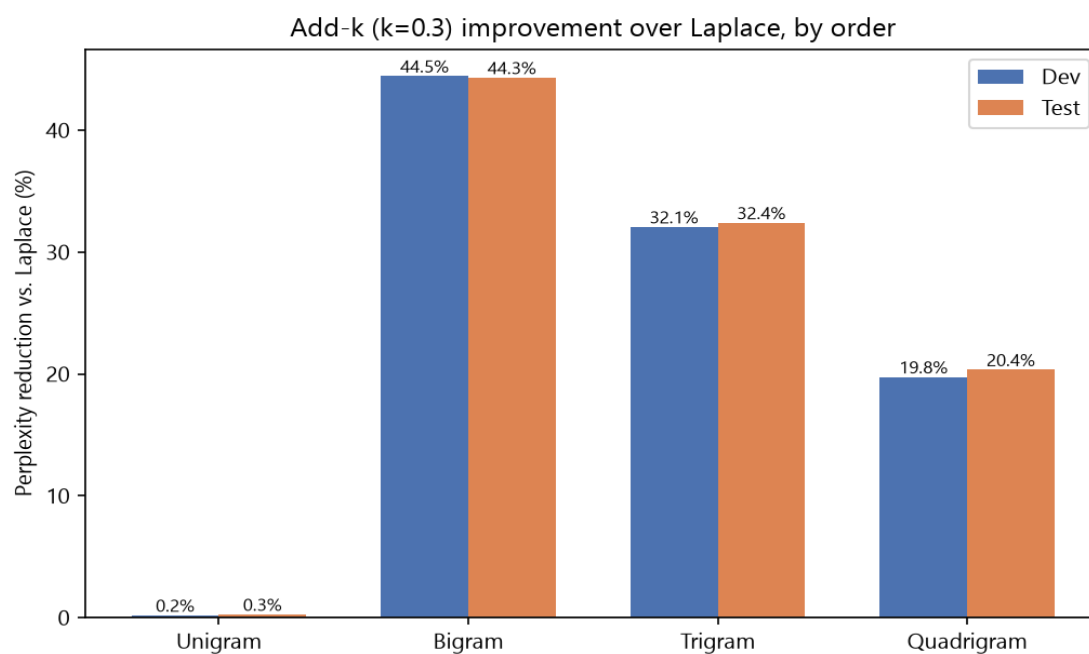


Figure 3 — Percent perplexity reduction of Add-k over Laplace, by order.

4. Analysis

The size of the improvement is a story about the smoothing constant relative to V , not about k in isolation. The correction term subtracted from every seen n -gram's share of probability mass scales with $k \cdot V$; since $V = 222,199$, Laplace's $k=1$ term dwarfs typical observed counts from the bigram order upward, so most probability mass ends up redistributed onto n -grams that were never seen. Shrinking k to 0.3 cuts that term to 30% of Laplace's, which is why the largest relative gain (about 44%) appears at the **bigram** order, where counts are populated enough for the smoothing constant to be the dominant source of distortion.

The **unigram** order barely changes ($<0.5\%$) because there is no context to sparsify — the denominator is just the total token count ($\sim 16.2\text{M}$), against which even Laplace's $k \cdot V \approx 222,199$ is negligible. The **trigram** and **quadrigram** orders improve less than the bigram in relative terms because at those orders almost every test n -gram is itself unseen in training regardless of k — both models remain dominated by the same "probability mass spread across an astronomically large unseen n -gram space" failure mode Assignment 4 diagnosed for Laplace smoothing. Add- k with $k=0.3$ is a smaller version of that over-smoothing, not a fix for it: perplexity still climbs with order under both schemes, since neither backs off to a lower-order estimate for n -grams it has never seen.

A method like Good-Turing, Kneser-Ney, or interpolation/backoff — which redistribute mass based on how *often* low-count events occur rather than a flat additive constant — would be expected to improve, rather than degrade, as order increases.

5. Conclusion

Add- k smoothing with $k=0.3$ consistently outperforms Add-one (Laplace) smoothing on this Gujarati corpus across all four n -gram orders and both held-out splits, confirming that the smoothing constant is a meaningful hyperparameter rather than an arbitrary implementation detail. The improvement is largest where counts are dense enough for the constant to dominate the estimate (bigram) and smallest where it barely matters (unigram) or where sparsity overwhelms it regardless of k (quadrigram).